Type MS · Draw or label the parts of a synovial joint

Answer 1 of 5 · Question 7 part b · 6 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

7 (a) (i)		aerobic respiration (1)		1
	(ii)	• shorter (1) • fatter/wider/thicker (1)		2
	(iii)	• shortening causes tension/pull (1) • tendon inelastic (1) • pulls on radius/bone (1)		3
(b)		• ball and socket indicated labelled bone (1) • capsule/ligament labelled (1) • cartilage labelled (1) • synovial fluid labelled (1) • synovial membrane labelled (1) • overall appearance of diagram (1)		6

Total 12 marks

Type MS · Draw or label the parts of a synovial joint

Answer 1 of 5 · Question 7 part b · 6 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw or label the parts of a synovial joint

Answer 2 of 5 · Question 2 part b.iii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

2(a)	A; (1 and 3) (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(i)	D spongy bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	B compact bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(iii)	cartilage;		**(1)**

Question Number	Answer	Additional guidance	**Mark**
2(b)(iv)	shock absorber/cushions; allows free movement of joint/prevents grinding/reduces friction;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)	• Ca^{2+}needed; • forms part of bone structure / provides strength to bone/no calcium bones are less dense/weaker; • protein required; • for muscle development/component • of muscle; • vitamin D needed; • to allow uptake of Ca^{2+};		**(6)**

Total 12 marks

Type MS · Draw or label the parts of a synovial joint

Answer 2 of 5 · Question 2 part b.iii · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Draw or label the parts of a synovial joint

Answer 3 of 5 · Question 6 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

6 (a)	• where two bones; • come together/meet;		2
(b) (i)	• humerus; • ulna;		2
(ii)	• covers end (of ball) of humerus/L; • covers depression in ulna/M;		2
(c)	• movement not smooth/bones grind/rub together/less shock absorption/increased friction; • painful;		2
(d)	any four from • attached to end of muscle/attach muscle to bone; • inelastic/do not stretch; • muscle contracts; • pulls tendon; • causing M/ulna/bone to move;		4 Total 12

Type MS · Draw or label the parts of a synovial joint

Answer 3 of 5 · Question 6 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw or label the parts of a synovial joint

Answer 4 of 5 · Question 3 part a.i · 2 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number					
3	(a)	(i)	A = synovial fluid; B = synovial membrane;		2
		(ii)	• lubricate joint; • for easier movement/to reduce friction;		2
	(b)	(i)	• one section covering ball; • one section covering semi-circular part of bone; •		2
		(ii)	• movement not smooth/bones rub together/friction/less shock absorbing; • painful;		2
	(c)		• tendons attach to muscles; • attached to bone/lower arm; • muscle/biceps/triceps contracts; • generates pull on tendons/lower arm/bone; • tendons inelastic;	attaches muscles to bones = 2	5

Total : 13 marks

Type MS · Draw or label the parts of a synovial joint

Answer 4 of 5 · Question 3 part a.i · 2 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw or label the parts of a synovial joint

Answer 5 of 5 · Question 3 part b.i · 2 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
3	(a) (i)	A = synovial fluid; B = synovial membrane;		2
	(ii)	• lubricate joint; • for easier movement/to reduce friction;		2
	(b) (i)	• one section covering ball; • one section covering semi-circular part of bone; •		2
	(ii)	• movement not smooth/bones rub together/friction/less shock absorbing; • painful;		2
	(c)	• tendons attach to muscles; • attached to bone/lower arm; • muscle/biceps/triceps contracts; • generates pull on tendons/lower arm/bone; • tendons inelastic;	attaches muscles to bones = 2	5

Total : 13 marks

Type MS · Draw or label the parts of a synovial joint

Answer 5 of 5 · Question 3 part b.i · 2 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks